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A Scalable Framework for Precision-Boosted Healthcare Predictions with Advanced Algorithms

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***Abstract*— Healthcare data continues to grow in size and complexity. Hence, there's a rising need for more scalable and accurate prediction models. This study presents a framework that combines big data processing with advanced machine learning techniques to improve healthcare predictions, with diabetes mellitus as the primary use case. Apache Spark is used for data preprocessing, which distributes the large dataset to its worker nodes and completes the tasks at faster rate. This approach significantly reduces processing time when handling with millions of records. Most of the healthcare data often suffers from class imbalance, which is addressed with Synthetic Minority Over-sampling Technique (SMOTE) to develop a well-learned model. A StackingClassifier is built with three powerful boosting algorithms: XGBoost, CatBoost, and LightGBM. These models are combined to capture the strength of each algorithm and then enhanced by an AdaBoost meta-learner to boost precision. Further, doing cross validation and tuning the model with hyperparameters showed impressive results. SHAP and LIME is used for explaining the model's predictions. Evaluating the model with standard metrics like accuracy, precision, recall, and AUC-ROC shows up 98% accuracy and precision. This high precision indicates fewer false positives, making the model especially useful for real-world healthcare applications.**

Keywords—healthcare analytics, bigdata, ensemble techniques, diabetes prediction, spark.

I. Introduction

Healthcare data grows faster and faster, and hence, there are potential problems in developing accurate and scalable predictive models. There are several healthcare systems that need better and faster predictions. The advent of machine learning algorithms has gained its importance in healthcare predictions. Rapidly growing medical data requires increasingly sophisticated and scalable machine learning techniques. A novel framework can set new standards in the field of healthcare. Despite having variety of medical data, the diabetes mellitus data is used as a use case to test the framework.

Diabetes is one of the chronic diseases seen worldwide. The International Diabetes Federation (IDF) predicts that by 2030, 643 million persons worldwide will have diabetes [1], [3]. Early detection and treatment are critical to avoiding acute complications such kidney failure, nerve damage, and heart disease.

Traditional approaches such as logistic regression and decision trees work well for smaller data sets. However, it seems that such approaches perform worse for larger and more complex data sets. Scalability, efficiency and precision are the challenges that attract the attention of researchers. Ensemble learning has proven to be a successful technique for overcoming these issues since it combines the benefits of numerous models [13]. Boosting algorithms like XGBoost, CatBoost, and LightGBM are powerful tools that improve weaker models at each stage, resulting in stronger models [4],[5],[6],[14]. However, ensemble models sometimes lead to significant computational costs when applied over large data sets because numerous models are combined together [8],[10].

Big data technologies such as Apache Spark offer a solution for processing large amounts of data in an incredibly short time. Apache Spark's capabilities for in-memory computing make it a useful tool for data preparation and real-time analytics [8]. Although it is increasingly used in finance and telecommunications, its application in healthcare, especially for large-scale predictive analytics, is still under-researched [9], [10]. This study fills this gap by integrating Spark's big data processing and advanced ensemble learning techniques using the example of diabetes prediction

In this study, a new approach for disease prediction is presented by combining Apache Spark with a stacking classifier consisting of three state-of-the-art boosting algorithms. In this study, XGBoost, CatBoost and LightGBM algorithms are used as they possess their own strengths [5],[6],[14]. These three algorithms are chosen for their inherent merits - XGBoost is able to reduce bias; CatBoost

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offers better performance on categorical data; LightGBM can work better with large datasets. It is hypothesised that by stacking, these models will leverage their combined capabilities to achieve better prediction accuracy.

This study presents the following main contributions: First, this work focuses on a scalable pipeline for healthcare prediction based on stacked ensemble boosting models [4], [5], [6], [14]. Second, the applicability of Apache Spark in processing large files of healthcare data is described [8], [9], [10].

To test the proposed framework, the diabetes dataset is taken as a use case. From cleanup to feature engineering and transformation tasks, everything is processed efficiently using Spark's distributed architecture [8]. This reduces the effort required to prepare the data for modelling, which usually proves to be a bottleneck with conventional methods. The integration of ensemble learning techniques with the big data processing offered by Spark is a major step forward in predictive analytics in healthcare.

II. LITERATURE SURVEY

Prediction of diabetes has been a challenging problem requiring substantial amounts of medical and computational research into those constructions that can efficiently handle large and complex datasets. Diabetes is a chronic condition that leads to severe consequences for health, which has led to an exponentially great amount of research into the applications of machine learning in medical data. Traditional models like logistic regression, decision trees, and SVMs are considered good alternatives for the problem of predicting diabetes, more so when applied to smaller datasets due to their simplicity and interpretability [1], [2]. However, it is clear that these models have certain obvious scalabilities along with predictive accuracy that goes down in proportion to size and complexity of the datasets [3].

According to Khan et al. [2], traditional ML methods, like decision trees and random forests, can be beneficial with small datasets but are generally less capable than more complex ensemble methods in the contexts of big and complex healthcare data. Logistic regression, used for decades because it is easy to interpret, does not predict outcomes as well in the context of high-dimensional data with complex, nonlinear interactions [3]. Thus, more scalable algorithms for dealing with complexities in health care data have been moved toward by researchers, especially ensemble learning methods in which multiple models are used together in order to deliver better performance.

One of the very commonly used ensemble learning techniques, especially boosting methods, has received considerable interest in recent times due to their superior performance in a large number of predictive tasks, including diabetes prediction. Different from AdaBoost, boosting algorithms function as a stage-wise process that constructs weak classifiers in order to classify incorrectly classified instances, thus improving the performance of the model as a whole [4]. While AdaBoost is highly successful in improving the outcome of the classification problem, it might not go well for categorical data and with large datasets and can prove to be computationally intensive [4].

To address these issues, more contemporary algorithms have been created, including CatBoost [5] and LightGBM [6]. Prokhorenkova et al. [5] proposed the CatBoost algorithm, which can handle categorical data efficiently without requiring extensive preprocessing steps. It uses ordered boosting to prevent overfitting or target leakage. On the other hand, LightGBM is introduced by Ke et al. [6] using leaf-wise tree growth strategy that significantly improves computational efficiency especially in large scale applications. These innovations have fitted both CatBoost and LightGBM particularly well for health analytics applications because the datasets are large and feature-rich.

While excellent individual boosting models exist with AdaBoost, CatBoost, and LightGBM, additional predictive accuracy can often be obtained from stacking a number of models using a StackingClassifier. Model stacking combines the best of different base learners to create an enhanced meta-model. Recent work by Shwartz-Ziv and Armon [7] demonstrates that well-constructed stacked models can even outperform deep learning approaches on specific tabular datasets commonly found in healthcare applications.

Stacking can be a very promising strategy to improve the accuracy of diabetes predictions by combining the bias-reducing capabilities of AdaBoost, the categorical data handling of CatBoost, and the computational efficiency of LightGBM. This is because the proposed approach takes advantage of each boosting model's strengths in health care applications: giving a more accurate and reliable prediction system [5], [6], [14].

Scalable data processing frameworks are required as health data become large and complex. Apache Spark, one of the most powerful tools for high performance in quick processing of large datasets, is considered for such applications; it is a distributed computing framework in contrast to disk-based systems and has in-memory processing capabilities that reduce the time involved in analyzing data drastically and therefore are ideally suited for big data applications [8].

This versatility in supporting both batch and real-time data processing makes Spark so popular in the teleservice and finance industries, though in the healthcare analytics it is just gaining momentum [8], [9]. For instance, Gunasekaran et al. [9] and Khawaja et al. [10] have effectively demonstrated Spark's efficiency in handling large-scale health care data for predictive analytics. Its application in diabetes prediction, especially in combination with advanced ensemble learning techniques such as stacking, seems to be in its infancy.

The computation issue of handling over a million records will be overcome using the benefit of distributed processing in Spark. Such data pre-processing and models could be trained more efficiently with even a much-reduced time to prepare data for machine learning tasks [8], [10].

Despite the advancements made in ensemble learning and processing big data, there are only few research works, which integrate the two techniques within large-scale healthcare

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applications. Most of the research studies either concentrate on boosting predictive performance by means of machine learning models or optimizing pipelines for processing data through big data technologies. While both approaches could be integrated to bring about these two complementary techniques in play addressing challenges within scalability, accuracy, and results in predicting diabetes, their full exploration has not been done.

This research fills this gap through the introduction of a new ensemble learning model formed by a combination of StackingClassifier with preprocessing data pipelines on Spark. To say, the scale, effectiveness, and accuracy in one solution capable of predicting diabetes in large-scale healthcare datasets expands the envelope of existing predictive analytics in healthcare [8], [14].

III. METHODOLOGY

A. Overview

This study provides a comprehensive methodology for healthcare prediction that starts with Apache Spark for quick data preparation and then progresses to developing an advanced ensemble model. Figure 1 depicts the process flow for this work. It begins with the ingestion and parallelization of raw data, which allows for speedier processing. Figure 1 shows the process flow of this work. It starts with the ingestion and parallelization of raw data, which enables faster processing. This approach includes several pre-processing steps: missing values, conversion of categorical variables into numerical formats, and feature assembling before splitting into training and test sets. To control the imbalance of the target classes in the training data, the SMOTE algorithm is applied. Three advanced ensemble algorithms are used as base learners, with AdaBoost acting as a meta-learner to boost the predictive model's performance. This will be fitted and cross-validated to achieve peak performance. The model's interpretability is assessed with the free source tools SHAP and LIME. The proposed framework combines the advantages of big data technologies and sound machine learning techniques. The platform demonstrates the significant potential for scalable and accurate diabetes prediction in healthcare applications.

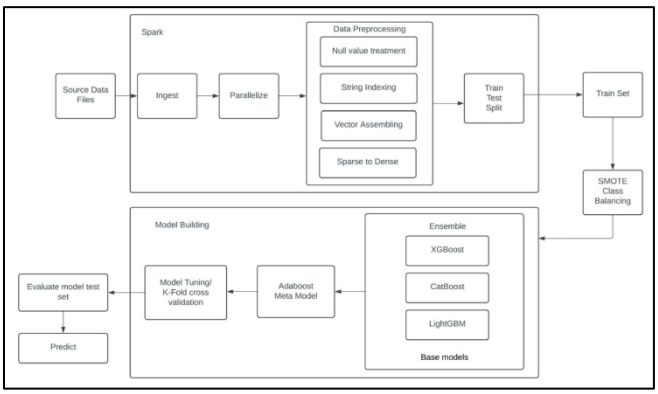

Figure 1. Overview

B. Dataset and Features

A publicly available diabetes dataset with 300,000 patient records with various medical attributes: Age (18-80 years), BMI, glucose levels, systolic and diastolic blood pressure, insulin levels, and family history of heart disease is used. The outcome variable is a binary variable indicating diabetes status as 1 for the case and 0 otherwise. The dataset is splited into a training set and a test set by performing stratified random sampling at a rate of 80% and 20%, respectively, to ensure class distribution.

C. Data Preprocessing Using Apache Spark

Apache Spark is used to preprocess the data. Spark's in-memory processing allows for concurrent execution of jobs across several processor cores. This considerably increases overall efficiency. The missing values are handled by Spark's imputer-transformer for continuous features. Mean value imputation was implemented.

The categorical attributes such as gender and smoking history are encoded into the numeric index using the String Indexing function so that the machine learning algorithms can interpret them correctly. For numerical attributes such as age, BMI, HbA1c and blood glucose, the features are transformed into mean and standard deviation using the Standard Scaling function. Normalizing the features helps the algorithm to learn faster during training.

Vector Assembler combines all features into a single vector and thus structures the data set for machine learning models. Thin vectors with a high proportion of zeros are also converted into the dense format. This converts them into an efficient format for calculation and makes processing smoother for the algorithms involved.

Finally, after conversion into a Pandas data frame, this dataset is ready to be fed into the machine learning pipeline with Scikit Learn. With these methods, the data is prepared well enough for modeling with Spark. It can be observed that Spark significantly shortens the calculation time, which is suitable for large-scale healthcare applications.

D. Feature Importance

A decision tree model is developed for the estimation of the feature importance in forecasting the outcome variable. From Figure 2, it is understood that the most important feature to infer the outcome variable is HbA1c Level, Blood glucose level, BMI level, and age. The age and smoking history have moderate importance. Heart disease and hypertension are the least significant variables and hence removed for model building as it could act as noise for learning.

E. Data Balancing Using SMOTE

After identifying the major characteristics, the dataset is split into training and testing by the 80:20 rule. Most of the medical dataset suffer from class imbalance. That is, the dataset will comprise only fewer positive records for disease.

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Because of this the learning will be biased which will finally affect the performance of the model. With no doubt, even the diabetes dataset also contains class imbalance; the positive diabetes cases are relatively fewer in proportion. To address this challenge, SMOTE is used to synthesize samples for the minority class to counter this imbalance. Unlike random over-sampling that duplicates minority instances, SMOTE creates new, unique samples by interpolating between existing minority samples. This approach ensures the synthetic samples are similar but not identical to the existing ones, enriching the dataset diversity. This will improve learning from the minority class, reducing the chances of overfitting and enabling better generalization to unseen data. The SMOTE method was applied exclusively to the training data, ensuring that the testing data remains representative of real-world scenarios. By balancing the training data, SMOTE enables the model to learn more effectively from the minority class, ultimately improving classification accuracy and model robustness.

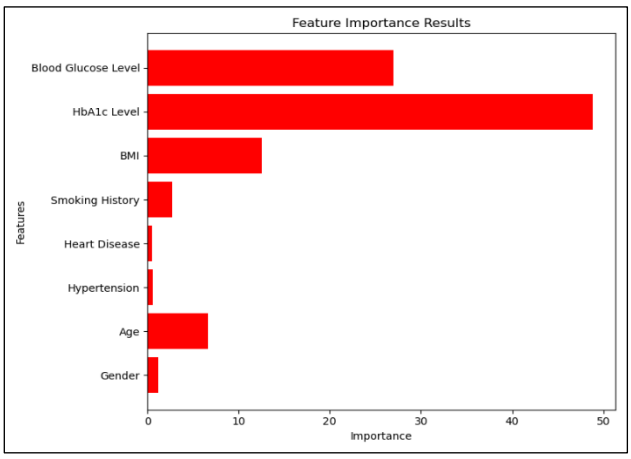

Figure 2. Feature Importance

F. Model Design

Classic machine learning algorithms such as Logistic Regression and KNN classification are attempted for building the model. To comprehensively evaluate the model's performance, Accuracy, Precision, Recall, F1 score and AUC-ROC scores are considered. The performance is measured and compared with classic ensemble techniques such as Random Forest classifier and AdaBoost classifier. An outperforming classic model is considered as the meta learner for the further stacking model.

The core model for prediction is a Stacking Classifier, which combines the predictions of multiple base models into a meta-model to improve accuracy. The Stacking Classifier uses the following models:

Base Learners

The base models used in this study are:

- XGBoost is a high-performance gradient boosting method designed for speed and accuracy. XGBoost's regularization aids in the prevention of overfitting.
- CatBoost is a gradient boosting algorithm that handles categorical data and eliminates overfitting through an ordered boosting technique.
- LightGBM is a highly efficient and scalable gradient boosting framework. LightGBM optimizes performance by leaf-wise growth, particularly for large datasets.

Meta-Learner

The predictions from the basic models are fed into the meta-model, which is an AdaBoost classifier. AdaBoost is well-suited for bias reduction since it focuses on the base models' faults while combining their strengths to improve final predictions. To increase performance, the meta-model's weights are frequently adjusted for misclassified data.

Hyperparameter Tuning

Grid Search with cross-validation was used to tune hyperparameters and optimize the performance of each base model and metamodel. The following hyperparameters were investigated.

- XGBoost parameters include learning rate (0.01, 0.05), maximum depth (4, 6, 8), and number of estimators (100, 200)
- CatBoost parameters include learning rate (0.01, 0.05), depth (6, 8, 10), and iteration count (1000).
- LightGBM parameters include number of leaves (31, 63), learning rate (0.01, 0.05), and iteration count (100, 200).

This stacking model was built with Python's scikit-learn module.

G. Model Explainability Using SHAP and LIME

To ensure trust, it is important to understand and interpret the machine learning model. It is specifically important to explain the intricacies of the model to the healthcare professionals. Especially, when complex ensemble models such as Stacking Classifier are deployed, it is important to explain the model. For better explainability, we used SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations).

LIME will enable us to obtain local explanations that provide visual insights on how individual features are affecting a specific prediction. This makes it much easier for the clinician to grasp why the model classified a patient as diabetic or not diabetic. Meanwhile, SHAP offers a global overview of how each feature affects all the predictions and gives us the complete picture about the behaviour of the model. By using both SHAP and LIME, critical insights are provided to the healthcare provider, facilitating real-world deployment and thereby enhancing patient care outcomes.

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IV. RESULTS AND DISCUSSION

This section is discussed in three folds: Data Preprocessing efficiency; Model Performance and the model explainability.

A. Impact of Apache Spark on Preprocessing Efficiency

Traditional preprocessing approaches would have led to bottlenecks in dealing with large datasets especially in terms of missing values and feature scaling. But using Apache Spark the preprocessing time has considerably improved as the architecture is based on distributed in-memory processing. Thus to measure the efficiency of Spark RDD (Resilient distributed dataset), the dataset is passed in an incremental fashion to measure the time taken by the process to handle each of the preprocessing steps. Figure 3 depicts the overall processing time of the framework, including staging jobs and collect the result back. Table I shows the break up of the time taken at various levels.

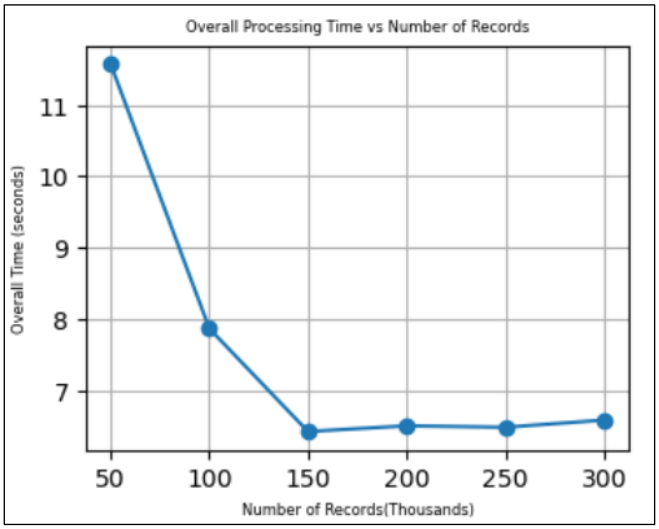

Figure 3. Overall Preprocessing Time

TABLE I. PREPROCESSING TIME FOR EACH STEP

Number of Records (K)	Preprocessing time (Secs) with varied record counts (K)				
	Null Value Treatment	String Indexing	Feature Assembly	Feature Scaling	Sparse to Dense
50	0.08	7.06	0.2	3.92	0.11
100	0.03	4.91	0.06	2.7	0.05
150	0.02	3.97	0.06	2.26	0.04
200	0.01	3.94	0.06	2.39	0.03
250	0.01	3.82	0.05	2.5	0.03
300	0.02	3.89	0.04	2.53	0.03

B. Model Performance

The performance of classical machine learning models and classical ensemble techniques were evaluated. There is a drastic performance shift between the classical models and the ensemble techniques. Table II shows the results of different evaluation metrics for the classical machine learning models (Logistic regression, KNN classification) and the ensemble techniques (Random Forest Classifier, AdaBoost Classifier). Since Adaboost classifier has provided impressive results, this meta-learner was selected. The individual base models XGBoost, CatBoost, and LightGBM, as well as the final stacking classifier, are also examined in the test set, which accounts for 20% of the original dataset. The results for each base learner and the stacking classifier are summarised below in Table III. The AUC and precision scores of different models are substantiated in Figure 4 and 5 respectively.

From the evaluation metric results, it is observed that all the base models has given an accuracy of 0.97. The most important result to be interpreted is precision. The LightGBM has given the highest precision of 0.91, meaning that it can actually detect cases of diabetes and would not miss many true diabetes cases due to false positives. The precision of XGBoost and CatBoost was 0.86.

TABLE II. COMAPARISION OF CLASSIC MODELS

Performance Metric	Classic Models Vs Classic Ensemble [score ranges from 0 to 1]			
	Logistic Regression	KNN	Random Forest	AdaBoost
Accuracy	0.88	0.96	0.96	0.96
Precision	0.4	0.79	0.79	0.79
Recall	0.89	0.74	0.74	0.74
F1 Score	0.55	0.76	0.76	0.76
AUC ROC Score	0.88	0.86	0.86	0.97

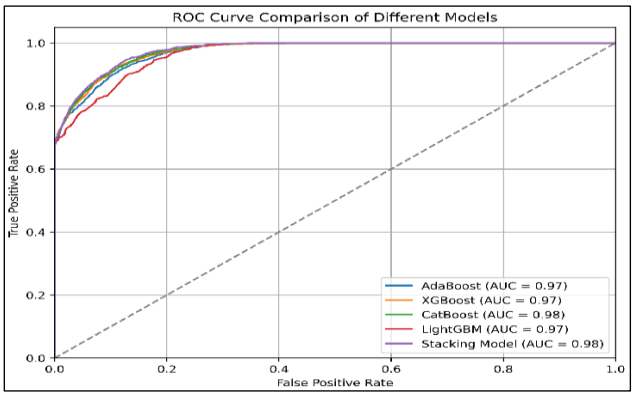

Figure 4. AUC Scores

TABLE III. COMPARISION OF INDIVIDUAL AND STACKING MODELS

Performance Metric	Performance of Advanced Algorithms & Stacking Classifier [score ranges from 0 to 1]			
	XGBoost	CatBoost	LightGBM	Stacking model

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Accuracy	0.97	0.97	0.97	0.97
Precision	0.86	0.86	0.91	0.97
Recall	0.72	0.72	0.71	0.71
F1 Score	0.79	0.79	0.8	0.8
AUC ROC Score	0.97	0.98	0.97	0.98

This separation clearly shows the importance of precision in diabetes prediction, where the correct identification of diabetes patients is crucial. The overall results of the experiment presented here show that the Stacking Classifier outperformed each of the models considering all of the evaluation metrics. Overall, the results obtained point to the importance of precision-oriented approaches in healthcare applications and show that LightGBM is a very efficient approach to make reliable predictions for diabetes.

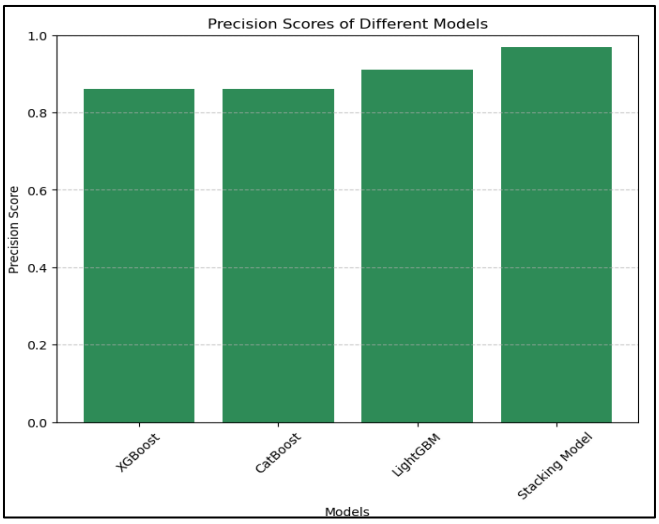

Figure 5. Precision Scores

C. *Model Interpretability and Clinical Relevance*

This explainability is vital for healthcare applications, because confidence in clinical settings depend on understanding the rationale behind model predictions. Although the machine learning models function as a black box, SHAP and LIME tools are used to describe the model's prediction in a more accessible fashion. SHAP provides the global understanding of the model while LIME provides the local insights. Figure 6 shows SHAP value while Figure 7 demonstrates the local interpretable results using LIME.

The SHAP value computed for each feature shows the contribution of that particular feature to its contribution in increasing or decreasing the values, with positive contributions indicated towards higher risks of diabetes and less is that of minus value, indicating lesser risks. In this case, as found in the models, the HbA1c level and blood glucose level is being the biggest predictors; whereas features such as gender and smoking history has smaller impacts.

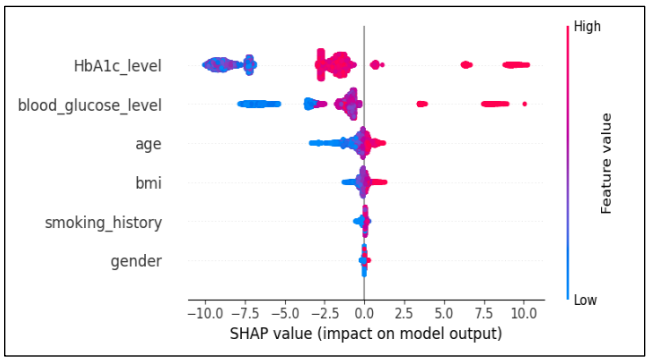

Figure 6. SHAP Value

This shows lesser value features are significant but not critical to the diagnosis of diabetes. This analysis supports the emphasis on blood glucose metrics in the clinical assessment of risk for diabetes. Interpretation of LIME outputs, The highest positive contribution to the prediction comes from HbA1c level ranging from 0.24 to 0.42. The second highest is the blood glucose level ranging from 0.27 to 0.36. As age more than 0.75 also contributes positively but the contribution is relatively less. BMI, smoking history, and gender appear to contribute very marginally as their importance values are almost zero.

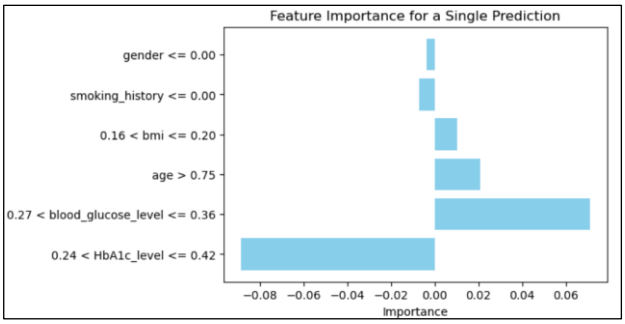

Figure 7. LIME result for Single Prediction

Using the SHAP and LIME, the Stacking Classifier is made more understandable and accessible to healthcare workers.

V. CONCLUSION AND FUTURE WORK

In this study, an efficient and scalable framework for health-care prediction is designed where it can boost the precision, taking diabetes prediction as a use case. The framework has Apache Spark for preprocessing data and a Stacking Classifier that combines XGBoost, CatBoost, and LightGBM, with AdaBoost as a meta-model. Using the distributed computing feature of Spark, the preprocessing time is drastically reduced and the ensemble model acheived high accuracy and precision. The use of SMOTE helped in the balance of the dataset. SHAP and LIME provided critical information about predictions by the model, which can be understandable to healthcare professionals. This approach has extreme potential for large-scale applications in healthcare, where speed and accuracy in prediction are of high importance. This model has to be tested on other healthcare datasets to check if it is generalizable beyond prediction of diabetes and determine whether it improves in future models. This approach can be added to real-time health

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care information systems, for example, electronic health records, and it might provide direct clinical insights to such patient care and early diagnosis.

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